

Technical Documentation

Secure Layer 2 Overlay Network via WireGuard

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Contents

1 Project Overview

1.1 Objective

Establish a secure, scalable Layer 2 (L2) network overlay between a VPS (Public Cloud) and a Proxmox Hypervisor (located behind CGNAT). This allows Virtual Machines on the Proxmox host to utilize the VPS as a gateway, effectively bypassing CGNAT restrictions and providing a centralized egress point.

1.2 Architecture Summary

- **Transport Layer:** WireGuard VPN (L3 Tunnel).
- **Overlay Layer:** GRETAP Tunnel (L2 Ethernet over IP).
- **Gateway:** VPS acts as the primary gateway and NAT point for the overlay subnet.

2 Network Topology & Addressing

2.1 Transport Network (WireGuard)

- **Subnet:** 10.77.77.0/24
- **VPS Endpoint:** <VPS_PUBLIC_IP>:51830
- **VPS Tunnel IP:** 10.77.77.1
- **Proxmox Tunnel IP:** 10.77.77.2

2.2 Overlay Network (L2 Bridge)

- **Subnet:** 10.50.0.0/24
- **VPS Bridge Gateway:** 10.50.0.1 (br-overlay)
- **Proxmox Bridge:** vmbr2 (No host IP assigned)
- **Test VM (AlmaLinux):** 10.50.0.10

3 Configuration: VPS (Gateway)

3.1 WireGuard Configuration

File: /etc/wireguard/wg-overlay.conf

```
1 [Interface]
2 Address = 10.77.77.1/24
3 ListenPort = 51830
4 PrivateKey = <VPS_PRIVATE_KEY>
5
6 # Enable forwarding for overlay
7 PostUp = sysctl -w net.ipv4.ip_forward=1
8 PostDown = sysctl -w net.ipv4.ip_forward=0
9
10 [Peer]
11 PublicKey = <PROXMOX_PUBLIC_KEY>
12 AllowedIPs = 10.77.77.2/32
```

3.2 Firewall & NAT

Requirements:

1. Open UDP port 51830.
2. Enable NAT (Masquerade) for the overlay subnet.

Commands:

```

1 # Open WireGuard Port
2 iptables -I INPUT -p udp --dport 51830 -j ACCEPT
3
4 # NAT for Overlay Subnet (Egress)
5 iptables -t nat -A POSTROUTING -s 10.50.0.0/24 -o ens1 -j MASQUERADE
6
7 # Save rules (Debian/Ubuntu)
8 iptables-save > /etc/iptables/rules.v4

```

3.3 Persistence (Systemd Service)

Due to Netplan limitations with GRETAP parents, persistence is handled via a custom Systemd service.

Script: /usr/local/bin/overlay-vps.sh

```

1 #!/bin/bash
2 # Wait for WireGuard interface
3 until ip link show wg-overlay >/dev/null 2>&1; do sleep 2; done
4
5 # Create GRETAP interface
6 ip link show gretap-overlay >/dev/null 2>&1 || \
7 ip link add gretap-overlay type gretap remote 10.77.77.2 local 10.77.77.1 dev
   wg-overlay
8 ip link set gretap-overlay up
9
10 # Create Bridge and attach GRETAP
11 ip link show br-overlay >/dev/null 2>&1 || ip link add br-overlay type bridge
12 ip link set br-overlay up
13 ip link set gretap-overlay master br-overlay
14
15 # Assign Gateway IP
16 ip addr show br-overlay | grep 10.50.0.1 || ip addr add 10.50.0.1/24 dev br-
   overlay
17
18 # Enable Forwarding
19 sysctl -w net.ipv4.ip_forward=1

```

Service: /etc/systemd/system/overlay.service

```

1 [Unit]
2 Description=Overlay Network Setup (GRETAP + Bridge)
3 After=wg-quick@wg-overlay.service
4 Requires=wg-quick@wg-overlay.service
5
6 [Service]
7 Type=oneshot
8 ExecStart=/usr/local/bin/overlay-vps.sh
9 RemainAfterExit=yes
10
11 [Install]
12 WantedBy=multi-user.target

```

Activation:

```
1 chmod +x /usr/local/bin/overlay-vps.sh
2 systemctl enable wg-quick@wg-overlay
3 systemctl enable overlay.service
4 systemctl start overlay.service
```

4 Configuration: Proxmox (Hypervisor)

4.1 WireGuard Configuration

File: /etc/wireguard/wg-overlay.conf

```
1 [Interface]
2 Address = 10.77.77.2/24
3 PrivateKey = <PROXMOX_PRIVATE_KEY>
4
5 [Peer]
6 PublicKey = <VPS_PUBLIC_KEY>
7 Endpoint = <VPS_PUBLIC_IP>:51830
8 AllowedIPs = 10.77.77.1/32
9 PersistentKeepalive = 25
```

4.2 Network Interfaces

File: /etc/network/interfaces

Added bridge vmbr2 for the overlay. Note: Proxmox naming convention does not allow hyphens (e.g., vmbr-overlay is invalid).

```
1 auto vmbr2
2 iface vmbr2 inet manual
3     bridge-ports none
4     bridge-stp off
5     bridge-fd 0
6     # GRETAP is attached dynamically via script or manually
```

4.3 GRETAP Setup Script

Script: /usr/local/bin/overlay-proxmox.sh

```
1 #!/bin/bash
2 # Wait for WireGuard
3 until ip link show wg-overlay >/dev/null 2>&1; do sleep 2; done
4
5 # Create GRETAP
6 ip link show gretap-overlay >/dev/null 2>&1 || \
7 ip link add gretap-overlay type gretap remote 10.77.77.1 local 10.77.77.2 dev
8     wg-overlay
9 ip link set gretap-overlay up
10
11 # Attach to vmbr2
12 ip link set gretap-overlay master vmbr2
```

Activation:

```
1 chmod +x /usr/local/bin/overlay-proxmox.sh
2 systemctl enable wg-quick@wg-overlay
```

5 Verification & Troubleshooting

5.1 Verification Steps

1. **Check WireGuard Handshake:** Both sides should show a recent handshake.

```
1 wg show
2 # Output should include: "latest handshake: X seconds ago"
```

2. **Check Bridge Links:** Verify GRETA is enslaved to the bridge.

```
1 bridge link | grep gretap
2 # Output: gretap-overlay@wg-overlay ... master vmbr2 state forwarding
```

3. **Connectivity Test (From VM):** Ping the gateway and external internet.

```
1 ping 10.50.0.1      # Gateway (VPS Bridge IP)
2 ping 8.8.8.8        # External IP (Tests NAT)
3 ping google.com     # DNS Test
```

5.2 Key Issues Resolved

1. **Handshake Failure:** Caused by copying the Public Key into the Private Key field in the VPS configuration. Resolved by pasting the correct Private Key.
2. **Firewall Blocking:** `tcpdump` confirmed packets arriving but not processing. Resolved by correcting the Private Key mismatch.
3. **Proxmox Bridge Naming:** GUI rejected `vmbr-overlay` due to naming validation (no special characters). Resolved by renaming to `vmbr2`.
4. **Netplan Compatibility:** Netplan on the VPS did not support the `parent` directive for tunnels. Resolved by using a Systemd service script.

6 Final State Checklist

VPS: WireGuard enabled at boot.

VPS: GRETA/Bridge setup script enabled at boot.

VPS: IP Forwarding enabled (`sysctl`).

VPS: NAT (Masquerade) rule active.

Proxmox: WireGuard enabled at boot.

Proxmox: Bridge `vmbr2` created.

VM: Network interface bridged to `vmbr2`.

VM: Static IP configured in Overlay Subnet (`10.50.0.x`).